IMAGE RECEPTOR, PRINCIPLE OF IMAGE FORMATION AND PROCESSING OF RADIOGRAPHS

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LECTURE LEARNING OUTCOMES

- Classify the radiographic films, their structure and identify the uses of different radiographic films.
- Describe the composition of processing solutions, their functions and different processing techniques.
- Describe the principle of image formation

IMAGE RECEPTORS

- Types :

I. Analogue image receptors

1. Direct-action or Non-screen film [sometimes referred to as wrapped or packet film] This type of film is sensitive primarily to X-ray photons
2. Indirect-action or Screen film: used in combination with intensifying screens in a cassette. This type of film is sensitive primarily to light photons, which are emitted by the adjacent intensifying screens

green, blue

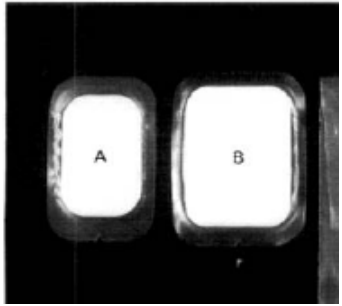
II. Digital image receptors

1. Solid state technology [CCD, CMOS,]
2. Photostimulable phosphor technology [Phosphor plate]

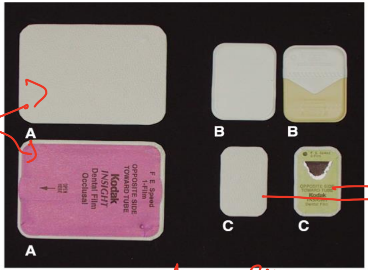
SIZES OF FILM

Intraoral film packets come in five basic sizes:

- Child [#0]
- Narrow anterior [#1]
- Adult size [#2] universal
- Preformed bitewing [#3]
- Occlusal [#4]



occlusal



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analoge film

STRUCTURE OF ANALOGUE RADIOGRAPHIC [DIRECT-ACTION] FILM

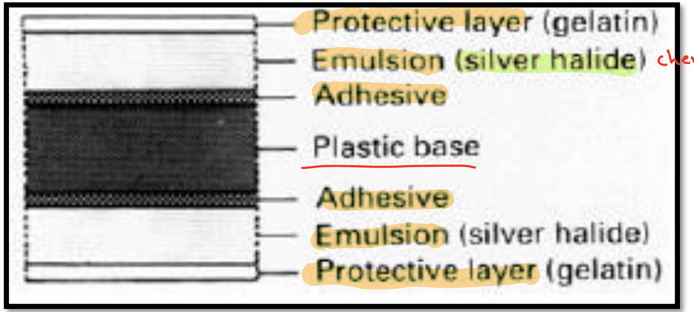
- The radiographic film comprises four basic components:

I. A Plastic Base: clear, transparent Cellulose Acetate which acts as a support for the emulsion but does not contribute to the final image

II. Adhesive: Fixes the Emulsion to the base because it's sensitive to radiation

III. Emulsion: On both sides of the base double

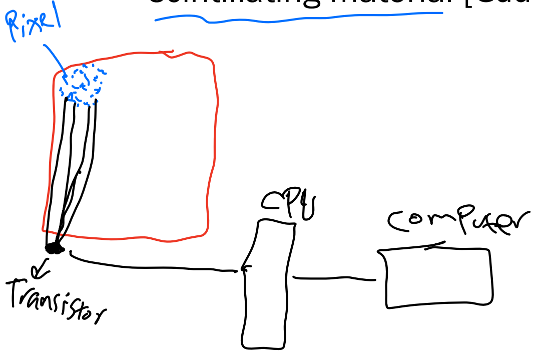
IV. A protective layer: Clear Gelatin to shield (protect) the emulsion from mechanical damage (ex. scratches)



chemical substance

DIGITAL IMAGE RECEPTORS
CHARGE-COUPLED DEVICE [CCD]

- CCD was introduced to dentistry in 1987, the first digital image receptor to be adapted for intraoral imaging.
- CCD uses a thin wafer of silicon crystals, which are formed in a pixel matrix Pecture element
- CCD detectors are more sensitive to light than to x-rays, therefore a layer of scintillating material [Gadolinium Oxybromide compounds] coated directly on the CCD



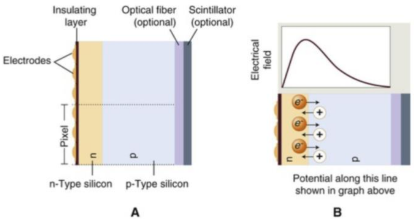
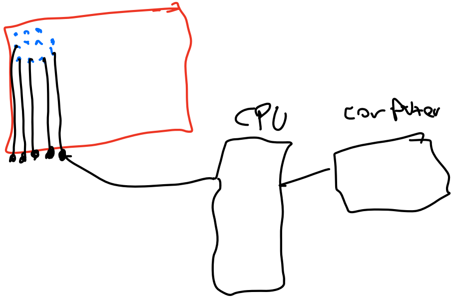


FIG. 4.4 (A) Basic structure of a charge-coupled device. The electrodes are insulated from an n-p silicon sandwich. The surface of the silicon typically incorporates a scintillating material to improve x-ray capture efficiency and fiberoptics to improve resolution. One pixel uses three electrodes. (B) Excess electrons from the n-type layer diffuse into the p-type layer, whereas excess holes in the p-type layer diffuse into the n-type layer. The resulting charge imbalance creates an electric field in the silicon with a maximum just inside the n-type layer.

DIGITAL IMAGE RECEPTORS CONT… COMPLEMENTARY METAL OXIDE SEMICONDUCTORS [CMOS]

- Are also silicon-based semiconductors
- Are fundamentally different from CCDs in the way that pixel charges are read
- Each pixel is isolated from its neighboring pixels and is directly connected to a transistor

direct



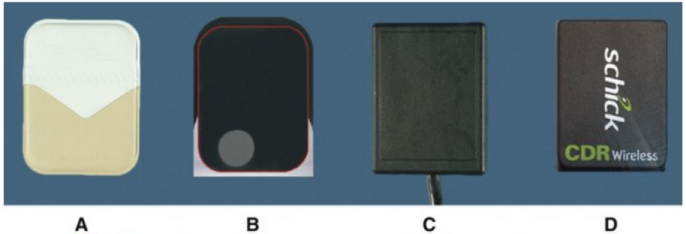


FIG. 4.7 (A) Kodak (courtesy Carestream Health, Inc., exclusive manufacturer of Kodak dental systems) No. 2 film. (B) Soredex (Milwaukee, WI) OpTime No. 2 PSP plate (outlined in *red)* placed on a barrier envelope to demonstrate packaged size. (C) Gendex (Hatfield, PA) No. 2 charge-coupled device sensor. (D) Schick (Sirona Dental Inc., Charlotte, NC) No. 2 complementary metal oxide semiconductor wireless sensor.

DIGITAL IMAGE RECEPTORS CONT… FLAT-PANEL DETECTORS

indirect

- use a photoconductor material selenium
- Flat-panel detectors are relatively expensive and currently limited to specialized imaging tasks, such as cone beam computed tomography (CBCT)

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DIGITAL IMAGE RECEPTORS CONT… PHOTOSTIMULABLE PHOSPHOR

indirect

- PSP plates absorb and store energy from x-rays and release this energy as light (phosphorescence) when stimulated by another light of an appropriate wavelength
- The PSP material used for radiographic imaging is "europium-doped" barium fluorohalide

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reuseble film

RADIOGRAPHIC FILM PROCESSING

- Analogue film processing
- Digital film processing

Xray
↓
film
↓
Form latent image
↓
Processing of image
↓ to be
visible image

PRINCIPLES OF IMAGE FORMATION

Interaction of X-rays with the film

Analogue film: chemical reaction [formation of latent image]	Digital film: electrically charged [formation of latent image]
Analogue film: chemical reaction with processing solution	Digital film: electrical charges are conveyed to the computer which are converted to digital signals
Analogue film: formation of image on the film	Digital film: formation of image on the computer screen

ANALOGUE FILM PROCESSING TECHNIQUES

- Manual or Wet processing
- Automatic processing
- Using self-developing films

MANUAL DEVELOPING

- Two Types:

1. Time Temperature Method
2. Visual Method

- Manual processing is usually carried out in a darkroom

- The requirements include:

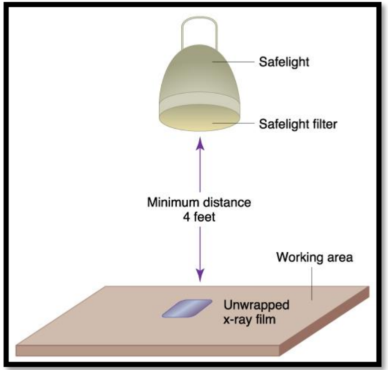
1. Absolute light-tightness
2. Adequate washing facilities
3. Infection control items [i.e. gloves, disinfectant, spray, and paper towels]

4. Container, labeled with a biohazard label, for contaminated film packets or barriers.

5. Recycle container for lead foil pieces; lead foil should not be thrown in the trash.

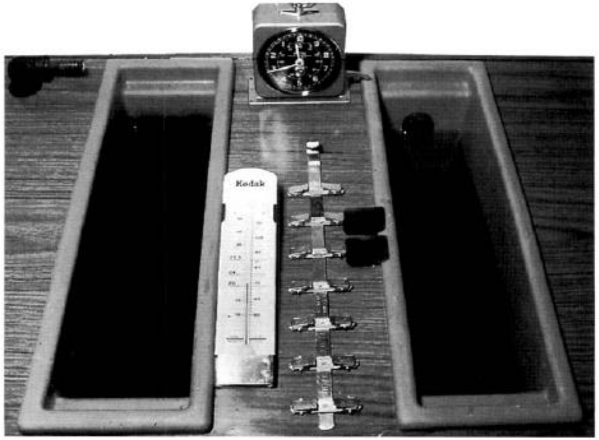
6. A source of white [normal] light

7. Safelights – positioned 1.2 m[4 feet] from the work surfaces with 25 W bulbs and filters suitable for the type of film



- Processing equipment:

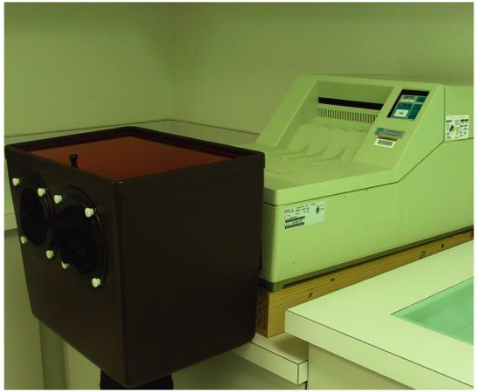
1. Tanks containing the various solutions
2. Thermometer
3. Immersion heater
4. An accurate timer
5. Film hangers



AUTOMATIC DEVELOPING

A

B



DIGITAL IMAGE PROCESSING

Any operation that acts to improve, restore, analyze, or in some way change a digital image is a form of image processing

- Image Restoration
- Image Enhancement
- Brightness and Contrast
- Sharpening and Smoothing
- Color

THANK YOU